Math Refresher for Masters

Homework 3

1 $S = \{v_1, v_2, v_3\} \subset \mathbb{R}^3$

$v_1 = (1, 2, 3)$
$v_2 = (2, 5, 8)$
$v_3 = (0, 1, 1)$

$$M = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 5 & 1 \\ 3 & 8 & 1 \end{pmatrix} \qquad \det M = 1\cdot5\cdot1 + 2\cdot1\cdot3 + 2\cdot8\cdot0 - 0\cdot5\cdot3 - 2\cdot2\cdot1 - 1\cdot1\cdot8 = 5+6+0+0-4-8 = -1 \neq 0$$

$\Rightarrow$ **S is linearly independent.**

2 $b_1 = (1, 0, 1)$
$b_2 = (0, 1, 1)$
$b_3 = (1, 1, 0)$

Verify: $\{b_1, b_2, b_3\}$ is a basis of $\mathbb{R}^3$.

$\Rightarrow$ must LI and span $\mathbb{R}^3$.

$$B = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \qquad \det B \overset{\text{by 1 row}}{=} 1\cdot\begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} + 0 + 1\cdot\begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = 1\cdot(-1) + 1\cdot(-1) = -2 \neq 0$$

$\Rightarrow$ **$\{b_1, b_2, b_3\}$ is a basis of $\mathbb{R}^3$.**

3

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

1) $u_1 = a_1 \qquad q_1 = \dfrac{u_1}{\|u_1\|} = \dfrac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

2) $u_2 = a_2 - \langle a_2, q_1 \rangle q_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \dfrac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}(1\ 1\ 0)\,\dfrac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \dfrac{1}{2}\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1/2 \\ -1/2 \\ 1 \end{pmatrix} \sim \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$

$q_2 = \dfrac{1}{\sqrt{1+1+4}}\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = \dfrac{1}{\sqrt{6}}\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$

3) $u_3 = a_3 - \langle a_3, q_2\rangle q_2 - \langle a_3, q_1\rangle q_1 =$

$$= \begin{pmatrix}0\\1\\1\end{pmatrix} - \frac{1}{\sqrt{6}}\cdot\frac{1}{\sqrt{6}}\begin{pmatrix}0\\1\\1\end{pmatrix}\begin{pmatrix}1 & -1 & 2\end{pmatrix}\begin{pmatrix}1\\-1\\2\end{pmatrix} - \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\1\end{pmatrix}\begin{pmatrix}1 & 1 & 0\end{pmatrix}\begin{pmatrix}1\\1\\0\end{pmatrix} =$$

$$= \begin{pmatrix}0\\1\\1\end{pmatrix} - \frac{1}{6}\begin{pmatrix}1\\-1\\2\end{pmatrix} - \frac{1}{2}\begin{pmatrix}1\\1\\0\end{pmatrix} = \begin{pmatrix}0-\frac{1}{6}-\frac{1}{2}\\1+\frac{1}{6}-\frac{1}{2}\\1-\frac{1}{3}-0\end{pmatrix} = \begin{pmatrix}-\frac{2}{3}\\\frac{2}{3}\\\frac{2}{3}\end{pmatrix} \sim \begin{pmatrix}-1\\1\\1\end{pmatrix}$$

$$q_3 = \frac{1}{\sqrt{3}}\begin{pmatrix}-1\\1\\1\end{pmatrix}$$

$$Q = \begin{pmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}}\\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}}\\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}}\end{pmatrix}$$

4) $R = Q^T A$, $r_{ij} = \langle q_i, a_j\rangle$

$$r_{11} = \langle q_1, a_1\rangle = \begin{pmatrix}\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\\0\end{pmatrix}\begin{pmatrix}1 & 1 & 0\end{pmatrix} = \sqrt{2} \qquad r_{21} = \langle q_2, a_1\rangle = \begin{pmatrix}\frac{1}{\sqrt{6}}\\-\frac{1}{\sqrt{6}}\\\frac{2}{\sqrt{6}}\end{pmatrix}\begin{pmatrix}1 & 1 & 0\end{pmatrix} = 0$$

$$r_{12} = \langle q_1, a_2\rangle = \begin{pmatrix}\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\\0\end{pmatrix}\begin{pmatrix}1 & 0 & 1\end{pmatrix} = \frac{1}{\sqrt{2}} \qquad r_{22} = \langle q_2, a_2\rangle = \begin{pmatrix}\frac{1}{\sqrt{6}}\\-\frac{1}{\sqrt{6}}\\\frac{2}{\sqrt{6}}\end{pmatrix}\begin{pmatrix}1 & 0 & 1\end{pmatrix} = \frac{3}{\sqrt{6}}$$

$$r_{13} \langle q_1, a_3\rangle = \begin{pmatrix}\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\\0\end{pmatrix}\begin{pmatrix}0 & 1 & 1\end{pmatrix} = \frac{1}{\sqrt{2}} \qquad r_{23} = \langle q_2, a_3\rangle = \begin{pmatrix}\frac{1}{\sqrt{6}}\\-\frac{1}{\sqrt{6}}\\\frac{2}{\sqrt{6}}\end{pmatrix}\begin{pmatrix}0 & 1 & 1\end{pmatrix} = \frac{1}{\sqrt{6}}$$

$$r_{31} = \langle q_3, a_1\rangle = \begin{pmatrix}-\frac{1}{\sqrt{3}}\\\frac{1}{\sqrt{3}}\\\frac{1}{\sqrt{3}}\end{pmatrix}\begin{pmatrix}1 & 1 & 0\end{pmatrix} = 0$$

$$r_{32} = \langle q_3, a_2\rangle = \begin{pmatrix}-\frac{1}{\sqrt{3}}\\\frac{1}{\sqrt{3}}\\\frac{1}{\sqrt{3}}\end{pmatrix}\begin{pmatrix}1 & 0 & 1\end{pmatrix} = 0$$

$$r_{33} = \langle q_3, a_3\rangle = \begin{pmatrix}-\frac{1}{\sqrt{3}}\\\frac{1}{\sqrt{3}}\\\frac{1}{\sqrt{3}}\end{pmatrix}\begin{pmatrix}0 & 1 & 1\end{pmatrix} = \frac{2}{\sqrt{3}}$$

$$R = \begin{pmatrix}\sqrt{2} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ 0 & \frac{3}{\sqrt{6}} & \frac{1}{\sqrt{6}}\\ 0 & 0 & \frac{2}{\sqrt{3}}\end{pmatrix}$$

$$Q^T Q = \begin{pmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0\\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}}\\ -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}}\end{pmatrix}\begin{pmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}}\\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}}\\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}}\end{pmatrix} = \begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix} = I \quad \Rightarrow \quad Q^T Q = I$$

4 $X = (3, -1) \in \mathbb{R}^2$

new basis: $w_1 = (1, 1)$ $w_2 = (2, -1)$

$$X = C_1 w_1 + C_2 w_2 \Rightarrow (3, -1) = C_1(1, 1) + C_2(2, -1)$$

$$\left(\begin{array}{cc|c} C_1 & 2C_2 & 3 \\ C_1 & -C_2 & -1 \end{array}\right) \xrightarrow{\text{II} - \text{I}} \left(\begin{array}{cc|c} C_1 & 2C_2 & 3 \\ 0 & -3C_2 & -4 \end{array}\right) \Rightarrow \begin{array}{l} C_2 = \frac{4}{3} \\ C_1 = 3 - 2C_2 = 3 - \frac{8}{3} = \frac{1}{3} \end{array}$$

$$X_w = \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = \begin{pmatrix} 1/3 \\ 4/3 \end{pmatrix}$$

5

$$\begin{cases} x + 2y - z = 3 \\ 2x - y + 3z = 7 \\ -x + y + 2z = 0 \end{cases}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & -1 & 3 & 7 \\ -1 & 1 & 2 & 0 \end{array}\right) \begin{array}{l} \text{II} - 2\cdot\text{I} \\ \text{III} + \text{I} \end{array} \rightarrow \left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -5 & 5 & 1 \\ 0 & 3 & 1 & 3 \end{array}\right) \begin{array}{l} \text{II} + 2\text{III} \\ \text{III} - 3\text{II} \end{array} \rightarrow \left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & 7 & 7 \\ 0 & 0 & -20 & -18 \end{array}\right)$$

$$z = \frac{18}{20} = \frac{9}{10}$$

$$y = 7 - 7z = 7 - \frac{63}{10} = \frac{7}{10}$$

$$x = 3 + z - 2y = 3 + \frac{9}{10} - \frac{14}{10} = \frac{25}{10}$$

Answer: $\left(\frac{25}{10}, \frac{7}{10}, \frac{9}{10}\right)$

6

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$$

1) $\det(A - \lambda I) = 0$

$$\begin{vmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{vmatrix} = (4-\lambda)(3-\lambda) - 2 = 12 - 7\lambda + \lambda^2 - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 2)(\lambda - 5) = 0 \Rightarrow \begin{bmatrix} \lambda_1 = 2 \\ \lambda_2 = 5 \end{bmatrix}$$

2) for $\lambda_1 = 2$: $(A - 2I)v_1 = 0$

$$\begin{pmatrix} 4-2 & 1 \\ 2 & 3-2 \end{pmatrix} v_1 = 0$$

$$\begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} v_1 = 0$$

$2v_x + v_y = 0$

$v_y = -2v_x \Rightarrow \boxed{v_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}}$

for $\lambda_2 = 5$: $(A - 5I)v_2 = 0$

$$\begin{pmatrix} 4-5 & 1 \\ 2 & 3-5 \end{pmatrix} v_2 = 0$$

$$\begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} v_2 = 0$$

$-v_x + v_y = 0$

$v_x = v_y \Rightarrow \boxed{v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}}$

3) $A = PDP^{-1}$

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix} \qquad D = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}$$

$$P^{-1} = \frac{1}{\det P} \begin{pmatrix} -2 & -1 \\ -1 & 1 \end{pmatrix} = \frac{1}{-3} \begin{pmatrix} -2 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{pmatrix}$$

$$\boxed{A = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{pmatrix}}$$

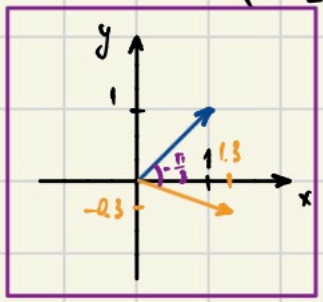
Check: $\begin{pmatrix} 5 & 2 \\ 5 & -4 \end{pmatrix} \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{pmatrix} = \begin{pmatrix} \frac{10}{3} + \frac{2}{3} & \frac{5}{3} - \frac{2}{3} \\ \frac{10}{3} - \frac{4}{3} & \frac{5}{3} + \frac{4}{3} \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} = A \Rightarrow$ done!

7 Build the transformation that rotates $\mathbb{R}^2$ by angle $-\pi/3$

a) $R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \Rightarrow R(-\frac{\pi}{3}) = \begin{pmatrix} 1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & 1/2 \end{pmatrix}$

b) $v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$$Rv = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \boxed{\begin{pmatrix} \frac{1+\sqrt{3}}{2} \\ \frac{1-\sqrt{3}}{2} \end{pmatrix}} \approx \begin{pmatrix} 1.3 \\ -0.3 \end{pmatrix}$$



8

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 2 & 1 \\ 2 & -1 & 0 \end{pmatrix}$$

Rules: 1. row substraction doesn't change det.
2. row swap changes sign.
3. multiply a row by c – det is multiplied by c.

$$U = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 2 & 1 \\ 2 & -1 & 0 \end{pmatrix} \xrightarrow[\text{III}-\text{I}]{\text{II}-2\cdot\text{I}} \begin{pmatrix} 2 & 1 & 1 \\ 0 & 0 & -1 \\ 0 & -2 & 1 \end{pmatrix} \xrightarrow[1\text{ swap}]{\text{II}\leftrightarrow\text{III}} \begin{pmatrix} 2 & 1 & 1 \\ 0 & -2 & 1 \\ 0 & 0 & -1 \end{pmatrix}$$

$\Rightarrow \det A = -\det U$

minus, because 1 swap

$$\det A = -\det U = -\left(2 \cdot (-2) \cdot (-1)\right) = \boxed{-4}$$